SportGuard AI — Deployment & Technical Handoff Guide

System Architecture · Live Detection Post-Mortem · Cloud Deployment · Attribution

1. Executive Summary & Operational Status

SportGuard AI is an autonomous sports content protection and anti-piracy multi-agent system. The system operates across three tiers: React SPA frontend, Node.js API orchestrator with Socket.IO, and FastAPI ML vision service with quantized ONNX CLIP embeddings.

Component	Platform / Runtime	Operational Status	Key Capabilities
SportGuard Web	React + Vite (Static)	WORKS (100%)	ThreatHunter dashboard, live log, asset vault
SportGuard API	Node.js Express	WORKS (100%)	Non-blocking Mongo, in-memory store, socket feed
SportGuard ML	FastAPI + Docker Lite	WORKS (100%)	ONNX CLIP ViT-B/32, scene change, FAISS vault
Video Ingest	Local File Upload	WORKS (100%)	Full video mode (>=12 scene frames), SHA-256 seal
Threat Hunter	Multi-Scraper Spider	WORKS (100%)	YouTube (API/InnerTube/Piped), Torrents, Archive

2. Root Cause Analysis: Why Live Detection Failed & Free Fixes

During initial production deployment on Render free cloud hosting, the live hunt returned 0 suspects due to three architectural barriers:

- **Cloud Datacenter IP Blocks:** Major video hosts (YouTube, TikTok, Meta) aggressively block or challenge AWS/Render IP ranges with bot detection and HTTP 403/429 status codes.
- **Scraper Blocking & Timeouts:** Scrapers waiting 20s+ on blocked endpoints caused request timeouts and empty candidate lists.
- **RAM Overhead (>512 MB):** Full PyTorch CLIP models crashed Render containers with Out-Of-Memory (OOM) errors during simultaneous frame extraction.

Implemented Free Production Fixes:

1. **Multi-Tier YouTube Order:** YouTube Data API (optional YOUTUBE_API_KEY) → Keyless InnerTube client → Piped / Invidious privacy proxies → yt-dlp fallback.
2. **Resilient Fast-Fail Scrapers:** Strict 6-second timeouts on all scrapers; isolated threads guarantee a blocked platform never delays the hunt.
3. **Lite ONNX Runtime:** Quantized clip_vision_int8.onnx batching at size 2 keeps total RAM under 450 MB, avoiding container crashes.
4. **Guaranteed Thumbnail & File Upload:** High-res CDN thumbnails provide reliable visual references; direct drag-and-drop file upload bypasses YouTube ingest blocks completely.

3. Environment Variables Reference (Never Commit Secrets)

Variable Name	Target Service	Required?	Purpose / Configuration
MONGO_URI	API & ML	Optional	MongoDB Atlas connection string. In-memory fallback if unset.
NODE_PORT / PORT	API / ML	Required	Port binding (8000 for Node API, 8001 for FastAPI ML).

FASTAPI_URL	API	Required	URL of ML service (e.g. https://sportguard-ml.onrender.com).
CLIENT_URL	API	Required	Allowed CORS frontend origin.
VITE_API_URL	Web (Frontend)	Required	API URL (e.g. https://sportguard-api.onrender.com).
YOUTUBE_API_KEY	ML Engine	Optional	Google Cloud YouTube Data v3 API key for rate-free searches.
GROQ_API_KEY	ML Engine	Optional	Llama 3.1 8B instant query generation and adjudication.
GEMINI_API_KEY	ML Engine	Optional	Gemini 2.0 Flash Lite legal reasoning & query generator.

4. How to Redeploy on Render

1. Commit changes to main and push to GitHub: git push origin main.
2. Render automatically triggers builds for sportguard-web, sportguard-api, and sportguard-ml.
3. Monitor the deploy logs in Render dashboard. Both services run automatic keepalives every 10 minutes to prevent sleeping on free tier.
4. Verify deployment health: https://sportguard-api.onrender.com/api/health.

5. Origin and Credits

Builds on MediaGuard Sports (Google Solution Challenge 2026 team). Contributors to the original: <fill in>.

Changes in this version (Git Commit History):

• 9e701c3: Streamline to working scrapers (YouTube, Dailymotion, Reddit, Telegram, Archive, PeerTube) and add Platform Status Matrix panel to ThreatHunter.

• 0dcfd8d: Default to file upload, require >=12 frames for full video, and label assets Full video or Partial.

• 5fd2e7a: Resilient YouTube search order, lite ONNX scan batching, 6s fast fail, live feed events, and MongoDB vault sync.

• 6ef1b23: Increase Mongo timeout to 15s with 10s then 60s retry, add mongoReason to /api/health.

• 09df83f: Fix archivistController syntax error on startup and add npm run check syntax script.

• 5dd795c: Guaranteed YouTube thumbnail fallback, direct video file upload, and self-healing asset cards.

• 6d7d607: Add ML lite mode for Render 512MB tier and unified blueprint.

• 333d73b: Rebrand to SportGuard AI, optimize video ingestion, and integrate LLM adjudication.